

A Diophantine Problem in EXP *

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Abstract

Let $f(x, y)$ be an arbitrary polynomial in $Z[x, y]$ and a an arbitrary integer. The decision problem of determining whether or not $\exists z \forall x \exists y f(x, y) = az$ true in Z is in EXP, which is both NP-hard and co-NP-hard.

1 Introduction

Given a polynomial $f(\bar{x})$, $\bar{x} = (x_1, \dots, x_n)$, with integer coefficients, many questions arise. For instance, is $f(\bar{x}) = 0$ solvable? Can an algorithm be found here which decides the solvability of an equation? This is the renowned Hilbert's tenth problem. Matijasevich, based on work done by Robinson, Davis, and Putnam, showed that this problem is undecidable if the number of variables n is sufficiently large [2]. So far, we know that over natural numbers $n = 9$ suffices to be undecidable [4]. For equations in one unknown, algorithms to decide its solvability over Z are taught in high school. But it may surprise non-specialists that to the best knowledge of contemporary mathematics a general algorithm is not known which decides the solvability over Z of polynomial equations in two unknowns. Currently, we only have algorithms which decide polynomial equations in two unknowns of special forms [1, 10, 17]. It is also known that the computational complexity of some special quadratic diophantine equations in two unknowns is NP-complete [9].

Given

a polynomial $f(x_1, \dots, x_m, y_1, \dots, y_n)$ over Z , whether or not for every integer $a_1, \dots, a_m$ in Z , $f(a_1, \dots, a_m, y_1, \dots, y_n) = 0$ is solvable in Z or Q is called the problem of diophantine equations with parameters [12]. A more detailed study on this topic may be

found in [13]. When $f(x_1, \dots, x_n, y)$ is a polynomial over Z with $n > 0$, the decision problem of deciding whether or not $\forall x_1, \dots, \forall x_n \exists y f(x_1, \dots, x_n, y) = 0$ is true in Z is co-NP-complete [15]. This decision problem is undecidable over N if the number of variables $x_1, \dots, x_n$ is greater than or equal to 9 [14]. However, the decision problem of deciding whether or not $\forall x \exists y f(x, y) = 0$ is true in N is co-NP-complete [15].

Over the set of natural numbers N , quantifier prefixes over diophantine equations are classified by their decidability [5]. We may see that many cases with only four quantifiers are undecidable. In this paper we study the problem of diophantine equations with three alternating quantifiers. Rojas [11] has shown that for average cases there are some cases with three quantifiers which are NP-complete. In this paper we show that, for the worst case, we have a diophantine decision problem which is in EXP, and is both NP-hard and co-NP-hard. More specifically, let $f(x, y)$ be an arbitrary polynomial in $Z[x, y]$ and a an arbitrary integer. The decision problem of determining whether or not $\exists z \forall x \exists y f(x, y) = az$ true in N is in EXP, which is both NP-hard and co-NP-hard. It is well-known that $\text{NP} \subseteq \text{EXP}$ and $\text{co-NP} \subseteq \text{EXP}$. Also, unless $\text{NP} = \text{co-NP}$, then NPC is not contained in co-NP, and co-NPC is

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not contained in NPC. To our current knowledge, this decision problem in EXP is the best possible result that can be obtained.

For simplicity, our results are stated for polynomials over Z . For any polynomial g over Q , let c be the least common multiple of the denominators of the coefficients of g . Then $h = cg$ is a polynomial over Z . With Euclidean Algorithm the number c can be found in polynomial time. Thus, the result in this paper can be extended to polynomials over Q .

2 Preliminaries

Next, we introduce the concept of “near solutions” for a multi-variate polynomial to solve the decision problem.

Definition. Let $f(x, y)$ be a polynomial over Q . The polynomial $g(x) \in Q[x]$ is a near solution of $f(x, y)$ if and only if there is a number c in Q such that $f(x, g(x)) = c$.

Theorem 2.1 *There is a polynomial time algorithm which, given $f(x, y)$ a polynomial over Q with degree s , determines whether or not there is a unique $g(x) \in Q[x]$, with its constant term equal to 0, such that $f(x, y) = \sum_{i=0}^s c_i(y - g(x))^i$ for some $c_i \in Q$. If this is the case, then the polynomial $g(x)$ can be found in polynomial time. If it is not the case, then in polynomial time the algorithm finds all polynomials $\{g_1(x), \dots, g_t(x)\}$, $t \leq s$, such that $g_i(x) \in Q[x]$ and $f(x, g_i(x)) \in Q$ for $1 \leq i \leq t$.*

Lemma 2.2 *Let $f(x) = \sum_{i=0}^n a_i x^i \in Z[x]$, and $|a| > \|f\|^3$. Then for any integer b , $f(a) = f(b)$ if and only if $|a| = |b|$ and $f(x)$ is an even function.*

3 Main Result

Now, we state the main result of this paper.

Theorem 3.1 *Let $f(x, y)$ be an arbitrary polynomial in $Z[x, y]$ and a an arbitrary integer. The decision problem of determining whether or not*

$$\exists z \forall x \exists y f(x, y) = az$$

true in N is in EXP, which is both NP-hard and co-NP-hard.

We first give the algorithm.

Input: Polynomial $f(x, y)$ in $Z[x, y]$ and a in N .

output: “Yes” if $\exists z \forall x \exists y f(x, y) = az$ is true in N , “No” if $\exists z \forall x \exists y f(x, y) = az$ is false in N .

We first handle some special cases. Assume that the x degree of $f(x, y)$ is equal to zero. The, our decision problem becomes deciding whether or not $\exists z \exists y f(y) = az$ is true in N . This can be decided as follows:

Check every natural number b , $0 \leq b < a$, whether $f(b) \equiv 0 \pmod{a}$ or not. If there is a number b , where $0 \leq b < a$, such that $f(b) \equiv 0 \pmod{a}$, then answer “yes”, otherwise, answer “no”.

Assume that the y degree of $f(x, y)$ is equal to zero. Then, our decision problem becomes deciding whether or not $\exists z \forall x f(x) = az$ is true in N . This is true if and only if $f(x) \equiv b$, i.e., the x degree of $f(x)$ is equal to zero, and b is a multiple of a .

If a is equal to zero, then our decision problem becomes deciding whether or not $\forall x \exists y f(x, y) = 0$ is true in N . This is the decision problem of Diophantine equations with parameter over N , which is co-NP-complete [15].

From now on, we assume that both the x and y degrees of $f(x, y)$ are not equal to zero and $a \neq 0$.

Step 1. Finding near solutions.

Find all the near solutions of $f(x, y)$ with the algorithm described in Theorem 2.1 and Lemma 2.2. Then, check the leading coefficient of each near solution whether it is positive or not. There are three possible cases.

Case 1. The polynomial $f(x, y)$ has no near solution with positive leading coefficient. Then, answer “no” and stop.

Case 2. The set $G = \{g_1(x), \dots, g_t(x)\}$, where $g_i(x)$ is a near solutions of $f(x, y)$ with positive leading coefficient, is a nonempty finite set. We call this the “Finite Case”. Then go to **Step 2**.

Case 3. There is a polynomial $h(x) \in Z[x]$ with positive leading coefficient such that $f(x, y) = \sum_{i=0}^m a_i(by - h(x))^i$ where a_i and $b > 0$ are integers. We call this the “Infinite Case”. Then go to **Step 3**.

Step 2. Finite Case.

Let $G = \{g_1(x), \dots, g_s(x)\}$, where $g_i(x)$ are with positive leading coefficients, and $f(x, g_i(x)) = c_i$ for some $c_i \in Q$. Partition G as follows: $\Gamma = \{G_1, \dots, G_t, G'\}$ is a partition of G ; $g_i(x)$ and $g_j(x)$ belong to the same set $G_k \in \Gamma$ if and only if there is a natural number γ such that $c_i = c_j = a \cdot \gamma$. Polynomial $g(x) \in G$ is in G' if and only if $f(x, g(x)) \neq a \cdot \gamma$ for any natural number γ . Let $\Delta = \{G_1, \dots, G_t\}$ and $t \leq s$.

2.1. If $\Delta = \emptyset$, then answer “no” and stop, otherwise, continue.

2.2. Assume that $\Delta = \{G_1, \dots, G_t\} \neq \emptyset$. Check whether or not there is a k , $1 \leq k \leq t$, such that for every $c \in N$ there exists a $g(x) \in G_k$ and $g(c) \in Z$. We do the following: For each k , $1 \leq k \leq s$, and each $g_{k,i}(x) \in G_k$ let $D_{k,i}$ be the least common multiple of the denominators of the coefficients of $g_{k,i}(x)$ and

$$h_{k,i}(x) = D_{k,i} \cdot g_{k,i}(x).$$

Let D_k be the maximum of these $D'_{k,i}$ s and $S_k = \{b \in N, 0 \leq b < D_k\}$. For every k and i

and every $b \in S_k$, check whether or not

$$h_{k,i}(b) \equiv 0 \pmod{D_{k,i}}.$$

Let $S = \{c_1, \dots, c_s\}$ be the set of natural numbers such that $c_k \in S$, where $1 \leq k \leq s \leq t$, if and only if there is a near solution $g(x) \in G_k$ such that $f(x, g(x)) = c_k$ and, for every $g(x) \in G_k$ and for every $a \in S_k$ there exists an i

$$h_{k,i}(b) \equiv 0 \pmod{D_{k,i}}.$$

If $S = \emptyset$, then answer “no” and stop, otherwise, continue.

2.3. Assume that $S = \{c_1, \dots, c_s\} \neq \emptyset$. Let $\Delta' = \{G_1, \dots, G_s\} \subseteq \Delta$ such that if $g(x) \in G_k$, $1 \leq k \leq s$, then $f(x, g(x)) = c_k$ where $c_k \in S$. Let

$$b = \text{Max}_{1 \leq k \leq s} \text{Max}_i(|h_{k,i}(x)|).$$

We then check whether or not for every $c \in S$ and every natural number d , $0 \leq d \leq b$, such that $f(d, y) = c$ is solvable in N . If there is a $c \in S$ such that for every natural number d , $0 \leq d \leq b$, $f(d, y) = c$ is solvable in N , then answer “yes”. Otherwise, answer “no” and stop.

Step 3. Infinite Case.

There is a polynomial $h(x) \in Z[x]$ with positive leading coefficient such that $f(x, y) = \sum_{i=0}^m a_i(by - h(x))^i$ where a_i and $b > 0$ are integers.

Find all the possible values of α , where $0 \leq \alpha < b$, and

$$\alpha \equiv h(a) \pmod{b}$$

for some a with $0 \leq a < b$. Let $A = \{\alpha_1, \dots, \alpha_m\}$ be the set of all these values. For every α , $\alpha \in A$, we want to check whether or not there is an integer β , $0 \leq \beta < a$, such that

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i \equiv 0 \pmod{a}.$$

3.1.1. Assume that $|A| = 1$ and let the only number in A be α . Let c be the GCD of a and b . Check whether or not

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i \equiv 0 \pmod{a}$$

for every integer β , $0 \leq \beta < a/c$. If there is no β such that

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i \equiv 0 \pmod{a},$$

then answer “no” and stop, otherwise, continue.

3.1.2. Suppose that there is a β such that

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i \equiv 0 \pmod{a}.$$

If $a_m > 0$, then answer “yes” and stop, otherwise, continue.

3.1.3. Suppose that $a_m < 0$. We look for the largest number β , $0 \leq \beta \leq \sum_{i=0}^m |a_i|$, such that

$$\exists z \left(\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i = az \right)$$

is true in N . We may start from the number $\beta = \sum_{i=0}^m |a_i|$ check whether

$$\exists z \left(\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i = az \right)$$

true in N or not. If for every number β , $0 \leq \beta \leq \sum_{i=0}^m |a_i|$,

$$\exists z \left(\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i = az \right)$$

is false in N , then answer “no” and stop, otherwise, continue.

3.1.4. As soon as we find the number β , such that

$$\exists z \left(\sum_{i=0}^m a_i(b \cdot \beta - \alpha)^i = az \right)$$

is true in N , we stop the checking and do the following. Find the minimum value of $h(x)$ in N and denote it h . This can be done by evaluate $h(d)$ for every d , $0 \leq d \leq \|h\|$, then do the comparison of these values. Compare $-h$ with β . if $-h \leq \beta$ then answer “yes”, otherwise, answer “no” and stop.

3.2.1. Assume that $|A| > 1$. $A = \{\alpha_1, \dots, \alpha_m\}$ has at least two different members. For every integer β , $0 \leq |\beta| \leq \|f\|^3$, check whether or not

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha_1)^i \equiv 0 \pmod{a}.$$

If there is no number β such that

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha_1)^i \equiv 0 \pmod{a},$$

then answer “no” and stop, otherwise, continue.

3.2.2. Assume that for some of the numbers β

$$\sum_{i=0}^m a_i(b \cdot \beta - \alpha_1)^i \equiv 0 \pmod{a}.$$

Let $\Gamma = \{\gamma_1, \dots, \gamma_r\}$ be the set such that $\gamma_j \in \Gamma \subseteq Z$ if and only if there is a β with $0 \leq |\beta| \leq \|f\|^3$ such that $\sum_{i=0}^m a_i(b \cdot \beta - \alpha_1)^i = a\gamma_j$. For each $\gamma \in \Gamma$, solve the equation (with variable x)

$$\sum_{i=0}^m a_i x^i - a\gamma = 0$$

and let S_γ be the solution set. For every element δ in S_γ compute the value of

$$\delta' \equiv -\delta \pmod{b}$$

with $0 \leq \delta' < b$, and let T_γ be the set such that $\delta' \in T_\gamma$ if and only if $\delta \in S_\gamma$. Let D be a subset of Γ such that $\gamma \in D$ if and only if $A \subseteq T_\gamma$. If $D = \emptyset$, then answer “no” and stop, otherwise, continue.

3.2.3. Assume that $D \neq \emptyset$. For every $\gamma \in D$ let N_γ be the smallest negative integer in S_γ and $N_\gamma = 0$ if every element in S_γ is nonnegative. For every $\gamma \in D$ and every integer x' , $0 \leq x' \leq ||h(x)|| + |N_\gamma|$, check whether or not $\exists y f(x', y) = a\gamma$ is true in N . If there is a $\gamma \in D$ for every integer x' , $0 \leq x' \leq ||h(x)|| + |N_\gamma|$, $\exists y f(x', y) = a\gamma$ is true in N , then answer “yes”, otherwise, answer “no” and stop.

COMPLEXITY

The NP-hardness follows from the decision problem of determining the solvability of quadratic diophantine equation in two unknown over N is NP-complete [9]. That is, for the two-variable quadratics of the form $\alpha x^2 + \beta y - \gamma = 0$, where α, β, γ are positive integers, the decision problem of solvability in N is NP-complete. Now, let the $f(x, y)$ of our decision problem be simply $f(x, y) = \alpha(y - x)^2 - \gamma$, and $\beta = -a$. This means that $f(x, y)$ is in Case 3.1 of our algorithm. We claim that $\alpha x^2 + \beta y - \gamma = 0$ is solvable in N if and only if

$$\exists z \forall x \exists y f(x, y) = az$$

is true in N . Clearly, if $\alpha x^2 + \beta y - \gamma = 0$ is solvable in N with solution (a, b) , then for any value of x let $y = x + a$, we obtain that $\alpha(y - x)^2 - \gamma = a \cdot b$. Thus,

$$\exists z \forall x \exists y f(x, y) = az$$

is true in N . Also, if $\exists z \forall x \exists y \alpha(y - x)^2 - \gamma = a \cdot z$ is true in N , then there is a b such that $\forall x \exists y \alpha(y - x)^2 - \gamma = a \cdot b$ is true in N . When $x = 0$, there is a natural number d such that $\alpha d^2 - \gamma = a \cdot b$. This implies that $\alpha x^2 + \beta y - \gamma = 0$ is solvable in N . Hence, solving two-variable quadratics of the form $\alpha x^2 + \beta y - \gamma = 0$ is a special case of our decision problem. Therefore, our decision problem is NP-hard.

The co-NP-hardness follows from the fact that the decision problem of diophantine equations with parameters over N is co-NP-complete [15]. That is, when $f(x, y)$ is an arbitrary polynomial over Z , the decision problem whether or not $\forall x \exists y f(x, y) = 0$ is true in N is co-NP-complete. The decision problem of diophantine equations with parameters is a special case of our decision problem, too. We simply take $a = 0$ in our decision problem. This completes our proof that the decision problem of determining whether or not

$$\exists z \forall x \exists y f(x, y) = az$$

is true in N is in EXP, which is both NP-hard and co-NP-hard.

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